

Introduction to Lists

1. A list in Python is a collection which is:
 - A) Unordered and immutable
 - B) Ordered and immutable
 - C) Ordered and mutable
 - D) Unordered and mutable
 2. Which of the following data types can a list contain?
 - A) Only integers
 - B) Only strings
 - C) Only lists
 - D) Elements of different data types
 3. Elements of a list are enclosed within:
 - A) Parentheses ()
 - B) Curly braces {}
 - C) Square brackets []
 - D) Angle brackets <>
 4. List elements are separated by:
 - A) Semicolon
 - B) Space
 - C) Colon
 - D) Comma
 5. Indexing of a list in Python starts from:
 - A) 1
 - B) 0
 - C) -1
 - D) Any number
 6. A list containing another list as an element is called:
 - A) Multidimensional list
 - B) Recursive list
 - C) Nested list
 - D) Hierarchical list
 7. Which of the following is a valid list?
 - A) (1,2,3)
 - B) {1,2,3}
 - C) [1,2,3]
 - D) <1,2,3>
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Accessing List Elements

8. Accessing list elements is similar to accessing:
 - A) Dictionary values
 - B) Tuple elements

- C) String characters
 - D) Set elements
9. What happens if we access an index that does not exist in a list?
- A) ValueError
 - B) TypeError
 - C) IndexError
 - D) NameError
10. Which index gives the last element of a list?
- A) 0
 - B) 1
 - C) -1
 - D) len(list)
11. If $n = \text{len}(\text{list1})$, the last element can be accessed using:
- A) `list1[n]`
 - B) `list1[n+1]`
 - C) `list1[n-1]`
 - D) `list1[-n-1]`
12. Negative indexing in a list accesses elements from:
- A) Beginning
 - B) Middle
 - C) Right end
 - D) Random position
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Mutability of Lists

13. Lists in Python are:
- A) Immutable
 - B) Mutable
 - C) Fixed size
 - D) Read-only
14. Changing an element of a list after creation is:
- A) Not allowed
 - B) Allowed
 - C) Causes error
 - D) Creates new list
15. Which operation shows mutability of a list?
- A) `list1 + list2`
 - B) `list1 * 2`
 - C) `list1[2] = 50`
 - D) `len(list1)`
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List Operations

Concatenation

16. The operator used to concatenate two lists is:
- A) *
 - B) +
 - C) %
 - D) &
17. After concatenation, the original lists:
- A) Are deleted
 - B) Are modified
 - C) Remain unchanged
 - D) Become nested
18. Concatenation requires both operands to be:
- A) Strings
 - B) Tuples
 - C) Lists
 - D) Numbers
19. Concatenating a list with a string results in:
- A) Empty list
 - B) Syntax error
 - C) TypeError
 - D) ValueError
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Repetition

20. The operator used to repeat a list is:
- A) +
 - B) **
 - C) *
 - D) //
21. Repetition creates:
- A) A new modified list
 - B) Multiple references
 - C) A nested list
 - D) A tuple
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Membership Operators

22. The in operator checks whether:
- A) An index exists
 - B) A value exists
 - C) A list exists
 - D) A function exists

23. The not in operator returns True when:

- A) Element is present
 - B) Element is absent
 - C) List is empty
 - D) Index is invalid
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Slicing

24. Slicing a list returns:

- A) A single element
- B) A new list
- C) A reference to list
- D) A tuple

25. If start index is greater than stop index, slicing returns:

- A) Error
- B) None
- C) Empty list
- D) Original list

26. Omitting both start and stop indices slices:

- A) No elements
- B) First element
- C) Entire list
- D) Last element

27. A negative step in slicing results in:

- A) Empty list
 - B) Error
 - C) Reverse traversal
 - D) IndexError
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Traversing a List

28. Traversing a list means:

- A) Sorting elements
- B) Accessing each element
- C) Deleting elements
- D) Copying list

29. Which loop is commonly used to traverse a list?

- A) for
- B) while
- C) both for and while
- D) do-while

30. The function range(len(list1)) is used to:

- A) Create a copy of list
- B) Generate valid indices

- C) Reverse list
 - D) Sort list
31. Which traversal accesses elements directly?
- A) Using index
 - B) Using slicing
 - C) Using for-each loop
 - D) Using pop()
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Built-in Functions & List Methods

General Functions

32. Which function returns number of elements in a list?
- A) size()
 - B) length()
 - C) count()
 - D) len()
33. list() function without arguments creates:
- A) None
 - B) Empty list
 - C) Error
 - D) Tuple
34. Passing a string to list() creates:
- A) Nested list
 - B) List of characters
 - C) Empty list
 - D) Error
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append() vs extend()

35. append() adds elements:
- A) At beginning
 - B) At given index
 - C) At end as a single element
 - D) One by one
36. append() when given a list results in:
- A) Merging lists
 - B) Nested list
 - C) Error
 - D) Sorted list
37. extend() adds:
- A) Whole list as one element
 - B) Each element individually

- C) Only first element
 - D) Only last element
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insert(), count(), index()

- 38. insert() requires:
 - A) One argument
 - B) Two arguments
 - C) Three arguments
 - D) No arguments
 - 39. count() returns:
 - A) Index of element
 - B) First occurrence
 - C) Number of occurrences
 - D) Boolean value
 - 40. index() returns:
 - A) Count of element
 - B) Last index
 - C) First index
 - D) Boolean
 - 41. index() raises error if element is:
 - A) Duplicate
 - B) Negative
 - C) Not present
 - D) Zero
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remove() and pop()

- 42. remove() deletes element by:
 - A) Index
 - B) Value
 - C) Position
 - D) Slice
- 43. remove() deletes:
 - A) All occurrences
 - B) Last occurrence
 - C) First occurrence
 - D) Random occurrence
- 44. pop() without argument removes:
 - A) First element
 - B) Middle element
 - C) Last element
 - D) Entire list

45. pop() returns:
- A) None
 - B) Index
 - C) Removed element
 - D) Boolean
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reverse(), sort(), sorted()

46. reverse() changes the list:
- A) By creating new list
 - B) In-place
 - C) Alphabetically
 - D) Numerically
47. sort() modifies the list:
- A) By creating copy
 - B) In-place
 - C) By slicing
 - D) By extending
48. sorted() returns:
- A) Same list sorted
 - B) New sorted list
 - C) Reverse list
 - D) None
49. sort(reverse=True) sorts list in:
- A) Ascending order
 - B) Random order
 - C) Descending order
 - D) Alphabetical order only
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min(), max(), sum()

50. min() returns:
- A) Largest element
 - B) Smallest element
 - C) Sum of elements
 - D) Average
51. sum() works on a list containing:
- A) Strings
 - B) Mixed data types
 - C) Numeric elements
 - D) Nested lists only
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Nested Lists

52. To access an element inside nested list, we use:

- A) Single index
- B) Slice
- C) Two indices
- D) pop()

53. list1[i][j] means:

- A) i and j are elements
- B) jth list of ith element
- C) ith list and jth element
- D) Two different lists

Copying Lists

54. list2 = list1 creates:

- A) Copy of list
- B) New list object
- C) Alias
- D) Empty list

55. Changing list1 also affects list2 because:

- A) Deep copy
- B) Shallow copy
- C) Same reference
- D) Indexing

56. Which method creates a new list using slicing?

- A) list1()
- B) list1[:]
- C) copy(list1)
- D) list(list1[:])

57. Which module provides copy() function?

- A) math
- B) sys
- C) copy
- D) os

List as Argument to Function

58. When a list is passed to a function, Python passes:

- A) Copy of list
- B) Value
- C) Reference
- D) Index

59. Changes made to list inside function are reflected because:

- A) Lists are immutable
- B) Lists are mutable

- C) Lists are global
- D) Lists are static

60. Assigning a new list to parameter inside function:

- A) Modifies original list
- B) Deletes original list
- C) Creates local copy
- D) Causes error

61. id() function is used to check:

- A) Value equality
- B) Reference equality
- C) Index
- D) Length

List Manipulation Programs

62. Menu-driven programs mainly use:

- A) Conditional statements
- B) Loops
- C) Functions
- D) All of the above

63. Deleting element by position uses:

- A) remove()
- B) del
- C) pop()
- D) clear()

64. Deleting element by value uses:

- A) pop()
- B) remove()
- C) del
- D) index()

65. Sorting without changing original list requires:

- A) sort()
- B) reverse()
- C) sorted()
- D) extend()

Output-Based & Logic MCQs

66. list1[::-2] accesses elements:

- A) Forward skipping one
- B) Reverse skipping one
- C) Only even indices
- D) Only odd indices

67. `del list1[3:]` deletes:
- A) First 3 elements
 - B) Elements from index 3 onward
 - C) Last element only
 - D) Entire list
68. `del list1[:5]` deletes:
- A) Elements from index 5
 - B) Elements till index 5
 - C) First five elements
 - D) Last five elements
69. `del list1[::2]` deletes:
- A) Odd index elements
 - B) Even index elements
 - C) All elements
 - D) No elements
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Conceptual MCQs

70. Which operation modifies list in-place?
- A) +
 - B) *
 - C) `sort()`
 - D) `sorted()`
71. Which of the following does NOT modify the list?
- A) `append()`
 - B) `remove()`
 - C) `sorted()`
 - D) `pop()`
72. A list can grow dynamically because it is:
- A) Immutable
 - B) Fixed size
 - C) Mutable
 - D) Static